

MOTORCYCLE BLACK BOX

Smart Accident Recording and
Vehicle Monitoring System for Motorcycles



INSTITUT TEKNOLOGI
SEPULUH NOPEMBER



OUR TEAM



INSTITUT TEKNOLOGI
SEPULUH NOPEMBER



Galuh Andar Siwi

Project Manager

Overall
Project

Team
Management



Sri Kusmiyati

Hardware Specialist

ESP32

MPU6050



Riri Desi Nofitri

Software Specialist

MQTT

AI



Fathika Afrine A

Quality Assurance

Test Drive

Data
Verification

BACKGROUND

! PERMASALAHAN

- ✦ Indonesia memiliki jumlah sepeda motor lebih dari 120 juta unit.
- ✦ Tingginya angka kecelakaan lalu lintas menyebabkan kesulitan investigasi penyebab kecelakaan.
- ✦ Tidak terjadi rekaman data kendaraan sebelum dan sesudah kecelakaan.
- ✦ Korban sering mengalami kesulitan memperoleh bukti klaim asuransi.
- ✦ Kendaraan roda dua umumnya belum memiliki sistem black box.

DAMPAK:

Sulit tentukan penyebab • Kerugian finansial • Data tidak terdokumentasi

120 Jt+

Sepeda motor di Indonesia

#1

Penyumbang kecelakaan terbesar

Minim

Sistem black box pada roda dua

PROBLEM STATEMENT

✗ EXISTING CONDITION

- ✗ Tidak ada data pitch dan roll kendaraan
- ✗ Tidak ada rekaman kondisi kendaraan
- ✗ Sulit mengetahui kronologi kejadian



✓ NEEDED SOLUTION

- ✓ Perekaman data kendaraan secara real-time
- ✓ Deteksi kecelakaan otomatis
- ✓ Penyimpanan data sebelum & sesudah kecelakaan

PROPOSED SOLUTION

MOTORCYCLE BLACK BOX

Sistem Monitoring berbasis IoT yang mampu:



Merekam data perjalanan kendaraan



Menyimpan data sensor ke cloud



Mendeteksi kecelakaan secara otomatis



Mengirimkan informasi kepada keluarga / pihak terkait

PRODUCT OVERVIEW



HARDWARE COMPONENTS



ESP32



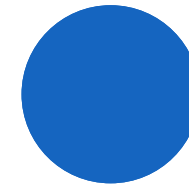
MPU6050 Accelerometer & Gyroscope



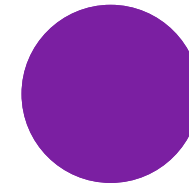
Power Management System



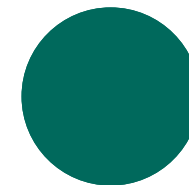
SOFTWARE COMPONENTS



Firmware ESP32

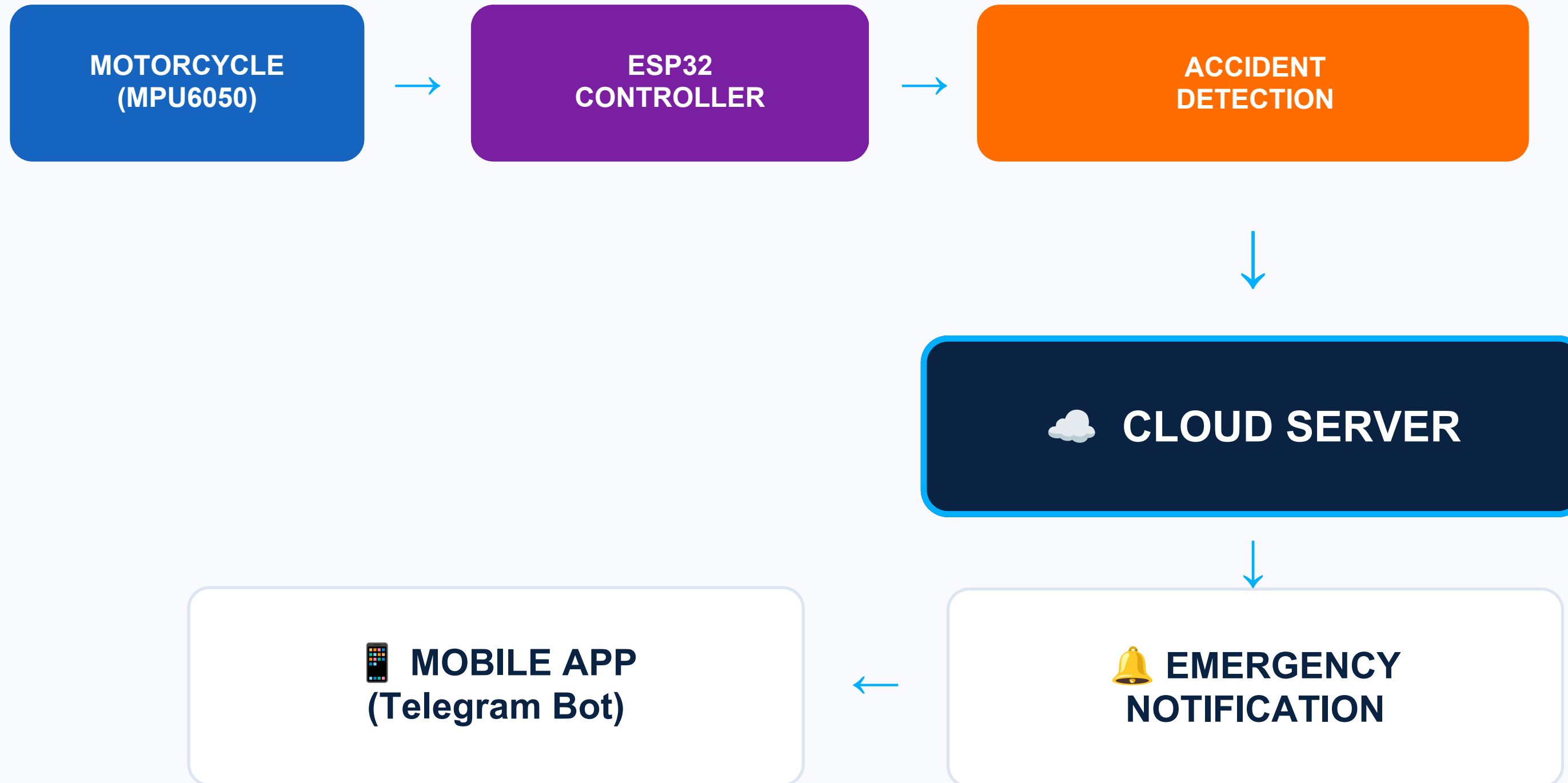


Firebase Database



Mobile Dashboard

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE



TEST DRIVE

insert video

TEST DRIVE

tampilan bot telegram

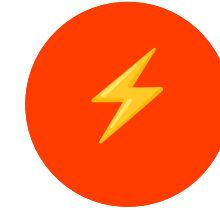
KEY FEATURES



REAL-TIME MONITORING

Pitch

Roll

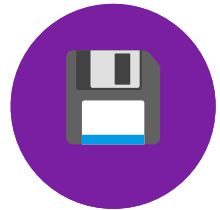


ACCIDENT DETECTION

Sudden impact detection

Vehicle rollover detection

Extreme acceleration detection



DATA LOGGING

Continuous Data Recording

Historical Tracking



EMERGENCY ALERT

Telegram Bot

INNOVATION & SWOT ANALYSIS

FITUR	Kendaraan Konvensional	Motorcycle Black Box
Rekam perjalanan	✗ Tidak ada	✓ Full trip recording
Bukti kecelakaan	✗ Tidak ada	✓ Available
Laporan darurat	✗ Manual	✓ Interactive based data
Riwayat rute	✗ Tidak ada	✓ Available

COMPETITIVE ADVANTAGE:

Low-cost solution

IoT-enabled

Easy installation

Real-time monitoring

Cloud connectivity

STRENGTHS

Affordable
Real-time monitoring
IoT based

WEAKNESSES

Requires internet connection

OPPORTUNITIES

Growing motorcycle market
Smart transportation trend
Insurance integration

THREATS

Competitors
Data security concerns

TARGET MARKET & BUSINESS MODEL

TARGET PASAR

PRIMARY MARKET

- Pengendara sepeda motor
- Driver ojek online
- Kurir pengiriman

SECONDARY MARKET

- Perusahaan asuransi
- Operator armada
- Instansi pemerintah

BUSINESS MODEL

1

HARDWARE SALES

Rp 500.000 – Rp 1.000.000/unit

2

SUBSCRIPTION SERVICE

Rp 10.000 – Rp 25.000/bulan

3

FLEET MONITORING SERVICE

Subscription for logistics companies

4

DATA ANALYTICS SERVICE

Traffic and safety analysis

COST ESTIMATION

KOMPONEN	BIAYA (Rp)
ESP32	80.000
MPU6050	20.000
Power Supply	35.000
Box Project	10.000
TOTAL	145.000

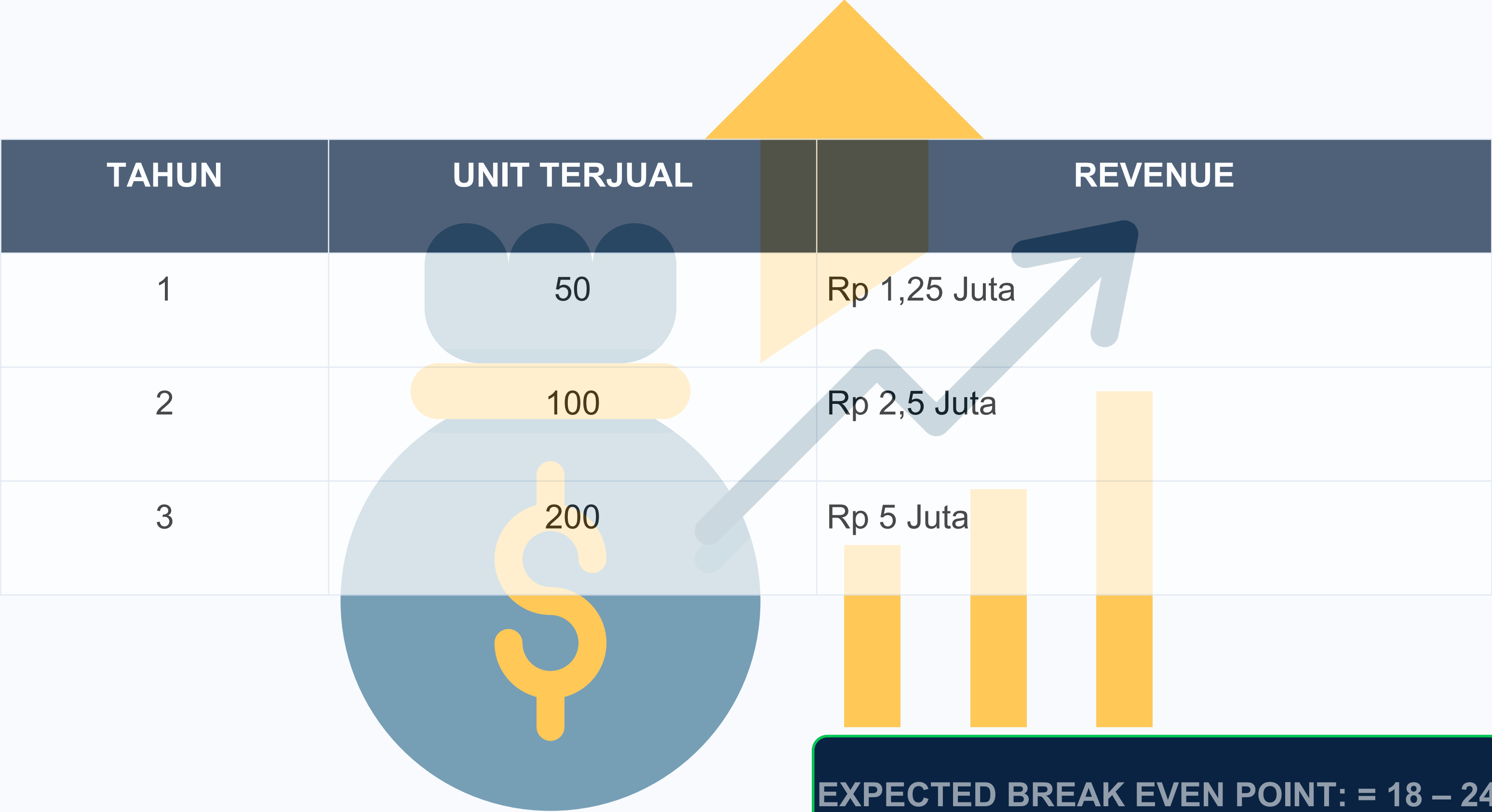
HARGA JUAL

Rp 250.000

ESTIMATED MARGIN

≈ 42%

FINANCIAL PROJECTION



DEVELOPMENT ROADMAP



SOCIAL IMPACT:



SAFETY

Mengurangi risiko keterlambatan penanganan korban kecelakaan.



LEGAL EVIDENCE

Menyediakan data objektif saat investigasi kecelakaan.



SMART TRANSPORTATION

Mendukung transformasi transportasi berbasis IoT.



ECONOMIC IMPACT

Membantu perusahaan asuransi dan operator armada mengurangi kerugian.

CONCLUSION

Motorcycle Black Box merupakan solusi IoT yang mampu:

- ✓ Merekam data kendaraan secara real-time
- ✓ Mendeteksi kecelakaan secara otomatis
- ✓ Menyediakan bukti digital kecelakaan
- ✓ Mengirim notifikasi darurat
- ✓ Mendukung peningkatan keselamatan pengendara sepeda motor

"Every Ride Recorded, Every Accident Explained."